

Chapter 2

Descriptive Statistics

GLOBAL
EDITION



Elementary Statistics

Picturing the World

EIGHTH EDITION

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Chapter Outline

- 2.1 Frequency Distributions and Their Graphs
- 2.2 More Graphs and Displays
- 2.3 Measures of Central Tendency
- 2.4 Measures of Variation
- 2.5 Measures of Position

Section 2.3

Measures of Central Tendency

Section 2.3 Objectives

- How to find the mean, median, and mode of a population and of a sample
- How to find the weighted mean of a data set, and how to estimate the sample mean of grouped data
- How to describe the shape of a distribution as symmetric, uniform, or skewed and how to compare the mean and median for each

Measures of Central Tendency

Measure of central tendency

- A value that represents a typical, or central, entry of a data set.
- Most common measures of central tendency:
 - Mean
 - Median
 - Mode



Measure of Central Tendency:

Mean

Mean (average)

- The sum of all the data entries divided by the number of entries.
- **Sigma notation:** Σx = add all of the data entries (x) in the data set.
- **Population mean:** $\mu = \frac{\Sigma x}{N}$
- **Sample mean:** $\bar{x} = \frac{\Sigma x}{n}$

Example: Finding a Sample Mean

The weights (in pounds) for a sample of adults before starting a weight-loss study are listed. What is the mean weight of the adults?

274 235 223 268 290 285 235



Solution: Finding a Sample Mean

274 235 223 268 290 285 235



- The sum of the weights is

$$\Sigma x = 274 + 235 + 223 + 268 + 290 + 285 + 235 = 1810$$

- To find the mean weight, divide the sum of the weights by the number of adults in the sample.

$$\bar{x} = \frac{\Sigma x}{n} = \frac{1810}{7} \approx 258.6$$

The mean weight of the adults is about 258.6 pounds.

Measure of Central Tendency:

Median

Median

- The value that lies in the middle of the data when the data set is **ordered**.
- Measures the center of an ordered data set by dividing it into two equal parts.
- If the data set has an
 - **odd number of entries:** median is the middle data entry.
 - **even number of entries:** median is the mean of the two middle data entries.

Example: Finding the Median

Find the median of the weight listed in the first example.

274 235 223 268 290 285 235



Solution: Finding the Median



- First, order the data.

223 235 235 268 274 285 290



- There are seven entries (an odd number), the median is the middle, or fourth, data entry.

The median weight of the adults is 268 pounds.

Example: Finding the Median

In the previous example, the adult weighing 285 pounds decides to not participate in the study. What is the median weight of the remaining adults?

223 235 235 268 274 290



Solution: Finding the Median



- First order the data.

223 235 235 268 274 290

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- There are six entries (an even number), the median is the mean of the two middle entries.

$$\text{Median} = \frac{235 + 268}{2} = 251.5$$

The median weight of the remaining adults is 251.5 pounds.

Measure of Central Tendency:

Mode

Mode

- The data entry that occurs with the greatest frequency.
- If no entry is repeated the data set has no mode.
- If two entries occur with the same greatest frequency, each entry is a mode (**bimodal**).

Example: Finding the Mode

Find the mode of the weights listed in Example 1.

223 235 235 268 274 285 290



Solution: Finding the Mode



- Ordering the data helps to find the mode.

223 235 235 268 274 285 290

- The entry of 235 occurs twice, whereas the other data entries occur only once.

The mode of the weights is 235 pounds.

Example: Finding the Mode

At a political debate a sample of audience members was asked to name the political party to which they belong. Their responses are shown in the table. What is the mode of the responses?

Political Party	Frequency, f
Democrat	46
Republican	34
Independent	39
Other/don't know	5



Solution: Finding the Mode

Political Party	Frequency, f
Democrat	46
Republican	34
Independent	39
Other/don't know	5



The response occurring with the greatest frequency is Democrat. So, the mode is Democrat. In this sample, there were more Democrats than people of any other single affiliation.

Comparing the Mean, Median, and Mode

- All three measures describe a typical entry of a data set.
- Advantage of using the mean:
 - The mean is a reliable measure because it takes into account every entry of a data set.
- Disadvantage of using the mean:
 - Greatly affected by **outliers** (a data entry that is far removed from the other entries in the data set).

Example: Comparing the Mean, Median, and Mode

The table shows the sample ages of students in a class. Find the mean, median, and mode of the ages. Are there any outliers? Which measure of central tendency best describes a typical entry of this data set?

Ages in a class						
20	20	20	20	20	20	21
21	21	21	22	22	22	23
23	23	23	24	24	65	

Solution: Comparing the Mean, Median, and Mode

Ages in a class						
20	20	20	20	20	20	21
21	21	21	22	22	22	23
23	23	23	24	24	65	

Mean: $\bar{x} = \frac{\Sigma x}{n} = \frac{20 + 20 + \dots + 24 + 65}{20} \approx 23.8 \text{ years}$

Median: $\frac{21 + 22}{2} = 21.5 \text{ years}$

Mode: 20 years (the entry occurring with the greatest frequency)

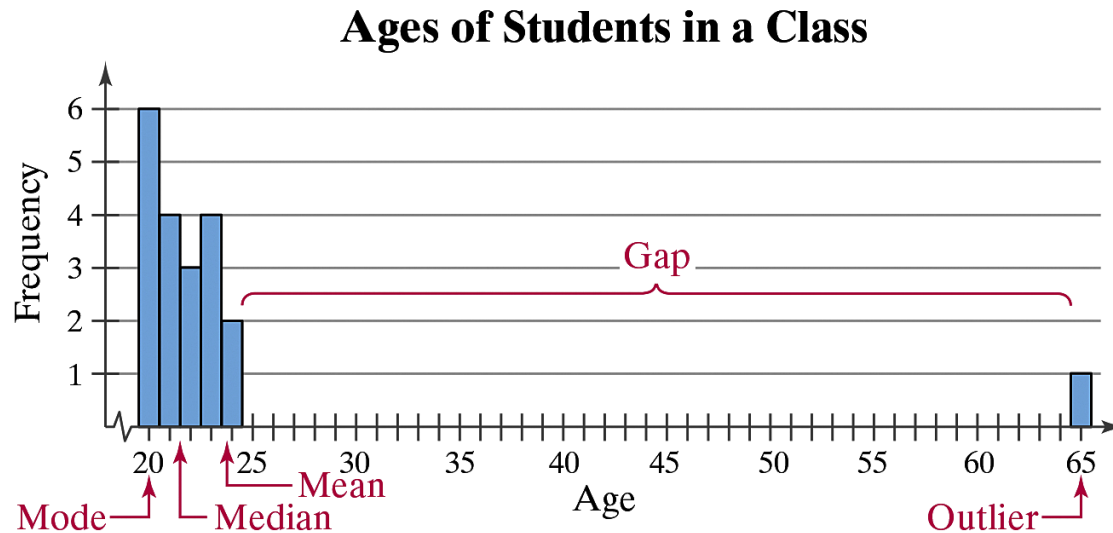
Solution: Comparing the Mean, Median, and Mode

Mean $\approx$ 23.8 years Median = 21.5 years Mode = 20 years

- The mean takes every entry into account, but is influenced by the **outlier** of 65.
- The median also takes every entry into account, and it is not affected by the outlier.
- In this case the mode exists, but it doesn't appear to represent a typical entry.

Solution: Comparing the Mean, Median, and Mode

Sometimes a graphical comparison can help you decide which measure of central tendency best represents a data set.



In this case, it appears that the **median** best describes the data set.

Weighted Mean

Weighted Mean

- The mean of a data set whose entries have varying weights.
- The weighted mean is given by

$$\bar{x} = \frac{\sum xw}{\sum w} \quad \text{where } w \text{ is the weight of each entry } x.$$

Example: Finding a Weighted Mean

Your grades from last semester are in the table. The grading system assigns points as follows: $A = 4$, $B = 3$, $C = 2$, $D = 1$, $F = 0$. Determine your grade point average (weighted mean).

Final Grade	Credit Hours
C	3
C	4
D	1
A	3
C	2
B	3

Solution: Finding a Weighted Mean

Points, x	Credit hours, w	xw
2	3	6
2	4	8
1	1	1
4	3	12
2	2	4
3	3	9
	$\Sigma w = 16$	$\Sigma (x \cdot w) = 40$

$$\bar{x} = \frac{\Sigma xw}{\Sigma w} = \frac{40}{16} = 2.5$$

Last semester, your grade point average was 2.5.

Mean of Grouped Data

Mean of a Frequency Distribution

- Approximated by

$$\bar{x} = \frac{\sum xf}{n} \quad n = \sum f$$

where x and f are the midpoints and frequencies of a class, respectively.

Finding the Mean of a Frequency Distribution

In Words

1. Find the midpoint of each class.
2. Find the sum of the products of the midpoints and the frequencies.
3. Find the sum of the frequencies.
4. Find the mean of the frequency distribution.

In Symbols

$$x = \frac{(\text{Lower limit}) + (\text{Upper limit})}{2}$$

$$\Sigma xf$$

$$n = \Sigma f$$

$$\bar{x} = \frac{\Sigma xf}{n}$$

Example: Find the Mean of a Frequency Distribution

The frequency distribution shows the cell phone screen times (in minutes) for 30 U.S. adults on a recent day. Use the frequency distribution to estimate the mean screen time. Using the sample mean formula, the mean screen time is 285.5 minutes. Compare this with the estimated mean.

Class midpoint, x	Frequency, f	xf
172.5	3	517.5
208.5	2	417.0
244.5	5	1222.5
280.5	6	1683.0
316.5	7	2215.5
352.5	4	1410.0
388.5	3	1165.5
	$n = 30$	$\Sigma = 8631$

Solution: Find the Mean of a Frequency Distribution

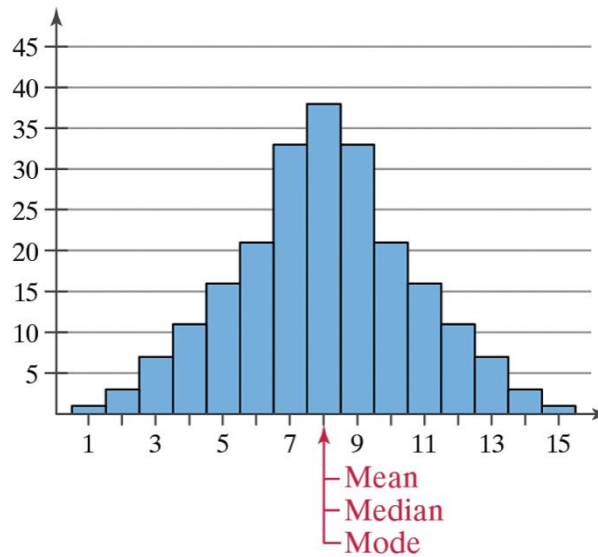
$$\bar{x} = \frac{\sum xf}{\sum n} = \frac{8631}{30} = 287.7$$

The mean screen time is 287.7 minutes. This value is an estimate because it is based on class midpoints instead of the original data set.

The Shape of Distributions

Symmetric Distribution

- A vertical line can be drawn through the middle of a graph of the distribution and the resulting halves are approximately mirror images.

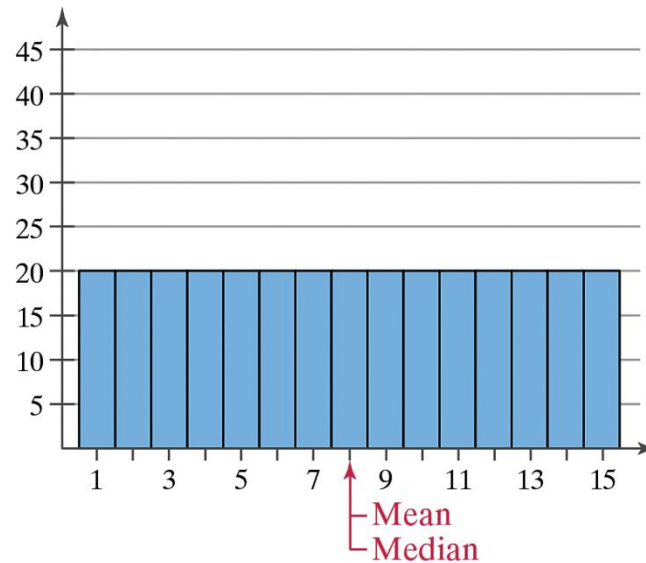


Symmetric Distribution

The Shape of Distributions

Uniform Distribution (rectangular)

- All entries or classes in the distribution have equal or approximately equal frequencies.
- Symmetric.

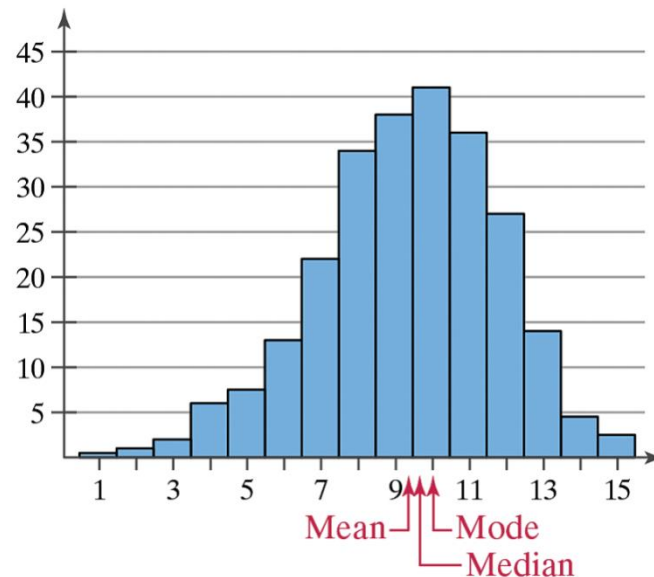


Uniform Distribution

The Shape of Distributions

Skewed Left Distribution (negatively skewed)

- The “tail” of the graph elongates more to the left.
- The mean is to the left of the median.

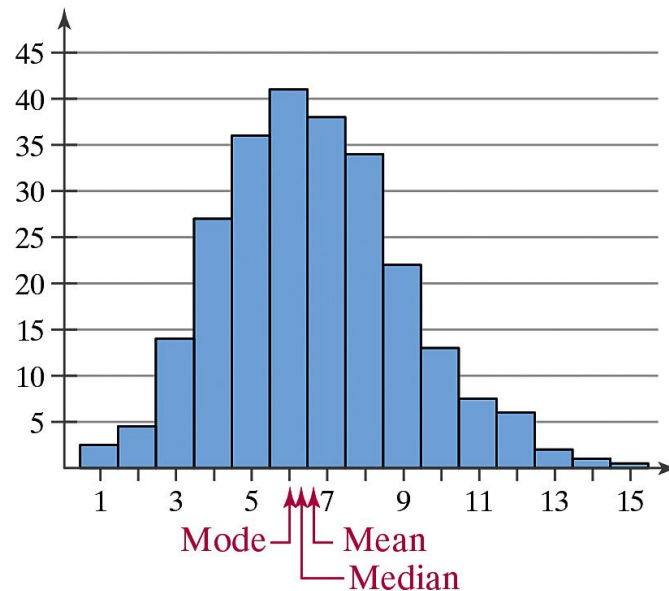


Skewed Left Distribution

The Shape of Distributions

Skewed Right Distribution (positively skewed)

- The “tail” of the graph elongates more to the right.
- The mean is to the right of the median.

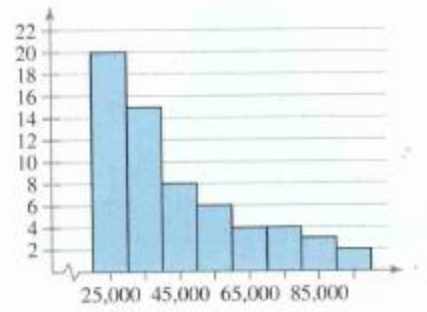


Skewed Right Distribution

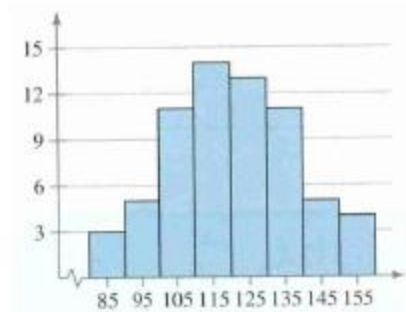
Exercise 1

Determine whether the approximate shape of the distribution in the histogram is symmetric, uniform, skewed left, skewed right, or none of these. Justify your answer.

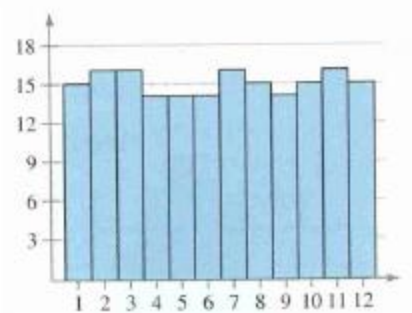
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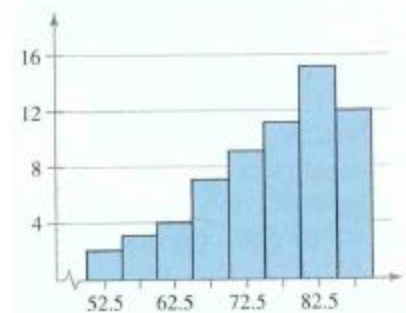
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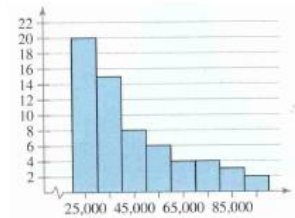


Exercise 2

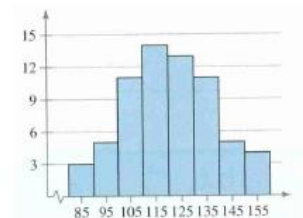
Match the distribution with one of the graphs in exercise 1. Justify your decision.

- The frequency distribution of 180 rolls of a dodecagon (a 12-sided die). Graph ____
- The frequency distribution of salaries at a company where a few executives make much higher salaries than the majority of employees. Graph ____
- The frequency distribution of scores on a 90-point test where a few students scored much lower than the majority of students. Graph ____
- The frequency distribution of weights for a sample of seventh grade boys. Graph ____

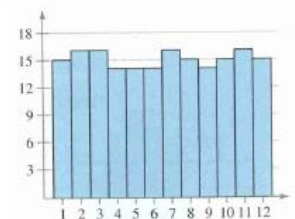
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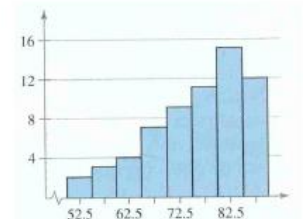
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3



4



Exercise 3

Find the mean, median, and mode of the data, if possible. If any of these measures cannot be found or a measure does not represent the center of the data, explain why.

Sports Cars The times (in seconds) for a sample of seven sports cars to go from 0 to 60 miles per hour.

3.7 4.0 4.8 4.8 4.8 4.8 5.1

Solution 3

Exercise 4

Find the mean, median, and mode of the data, if possible. If any of these measures cannot be found or a measure does not represent the center of the data, explain why.

Air Quality The responses of a sample of 1040 people who were asked if the air quality in their community is better or worse than it was 10 years ago.

Better: 346

Worse: 450

Same: 244

Solution 4

Exercise 5

Find the weighted mean of the data.

Final Grade The scores and their percents of the final grade for a statistics students are given. What is the student's mean score?

	Score	Percent of final grade
Homework	85	5%
Quizzes	80	35%
Project	100	20%
Speech	90	15%
Final Exam	93	25%

Solution 5

Exercise 6

Approximate the mean of the frequency distribution.

Ages The ages of residents of a town

Age	Frequency
0 – 9	55
10 – 19	70
20 – 29	35
30 – 39	56
40 – 49	74
50 – 59	42
60 – 69	38
70 – 79	17
80 – 89	10

Solution 6